

ECO::WATCH

Online Suspended Matter Sensor

User manual



User Notes

- Please read the instruction carefully before using and save it for reference.
- Please follow the instructions and precautions.
- When receiving the instrument, please open the packaging carefully, inspect equipment's damage level in case of transportation, if you found spoiled equipment, please immediately notify the manufacturer and distributor, and retain the packaging, in order to send back to processing.
- When the instrument is in trouble, please don't repair it by yourself, please directly contact the maintenance department of the manufacturer.

Content

User Notes	2
I.Operational principle	4
II.Technical performance and specifications.....	4
1. Technical parameters.....	4
2. Dimensional drawing	5
III.Installation and electrical connection.....	5
1. Installation.....	5
2. Electrical connection.....	6
IV.Maintenance And Maintenance	6
1. Maintenance procedures and methods	6
2. Sensor calibration.....	7
3. Frequently Asked Questions	7
V.Quality and service.....	7
1. Quality assurance	7
2. Accessories and spare parts.....	7
3. After-sales service commitment.....	8
Appendix data communication	8

I .Operational principle

ECO::WATCH TSS integrated on-line suspended matter sensor is designed and manufactured by the principle of scattered light suspended matter measurement method. When a beam is shot into the water sample, the light is scattered by the suspended matter in the water sample. By measuring the intensity of the backscattered light and comparing it with the internal calibration value, the suspended matter in the water sample can be calculated, and the final value can be output after linearization.

- scattered light principle, internal temperature sensor
- support RS - 485, Modbus/RTU protocol, 4-20 ma current output
- fiber structure, external light interference ability
- the infrared LED light source, high stability
- IP68 protection, within the depth of 20 meters

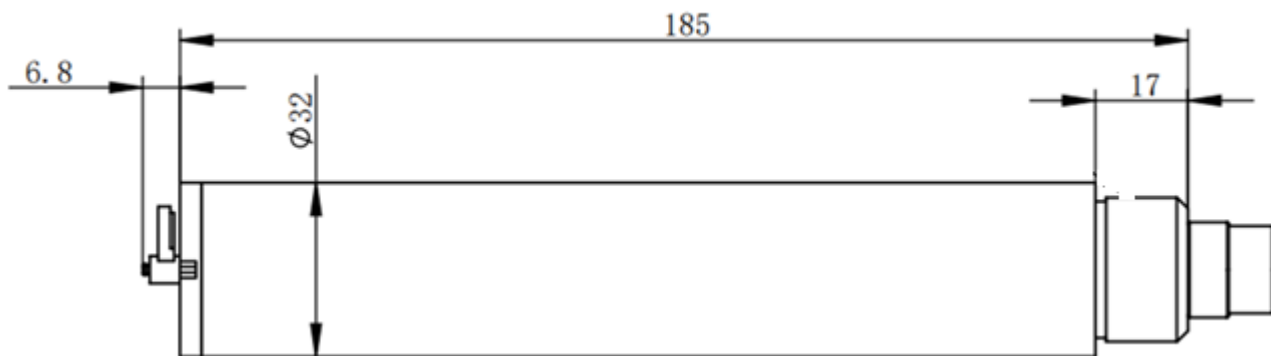
II. Technical performance and specifications

1. Technical parameters

Model	ECO::WATCH - TSS	
Case Material	316L	
Measurement Principle	Scattered light method	
Range and resolution	0~200.0 mg/L	0.1
	0~2000.0 mg/L	0.1
Accuracy	±5% (depending on sludge homogeneity), ±0.3°C	
Response time (T90)	<30s	
Minimum detection limit	1mg/L	
Calibration method	Two-point calibration	
Cleaning method	Built-in cleaning brush	
Temperature compensation	Automatic temperature compensation (Pt1000)	
Output method	RS-485 (Modbus RTU), 4-20 mA (optional)	
Storage temperature	-5~65°C	
Working conditions	0~60°C, ≤0.2MPa	
Installation method	Submersible installation, 3/4 NPT	

Power consumption	0.2W@12V
Power supply	12~24V DC
Protection level	IP68
Cable length	5 meters, other lengths can be customized

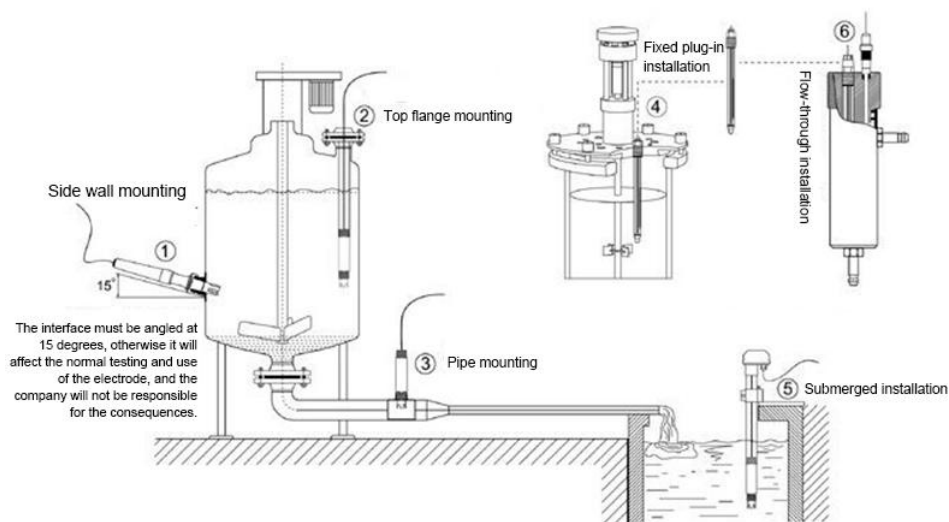
2. Dimensional drawing



Note: The sensor connector is a male M16-5-core waterproof connector.

III. Installation and electrical connection

1. Installation



Installation distance requirements: keep more than 5cm with the side wall and 10cm or more

with the bottom.

2. Electrical connection

- a) Red line - power cord (12 ~ 24V)
- b) Black line - ground (GND)
- c) Blue line - 485A
- d) white line - 485B
- e) Yellow line—current output (optional, can be left floating if not used)

After wiring is completed, it should be carefully checked to avoid incorrect connections before powering up.

Cable specification: Considering that the cable is immersed in water (including sea water) for a long time or exposed to the air, all the wiring points are required to do waterproof treatment, the user cable should have certain corrosion resistance.

IV. Maintenance And Maintenance

1. Maintenance procedures and methods

1.1 Maintenance schedule

The cleanliness of the measurement window is very important to maintain accurate readings.

Maintenance task	Recommended maintenance frequency
Calibrate the sensor (if required by the competent authority)	According to the maintenance schedule required by the competent authority

1.2 Maintenance methods

- Sensor outer surface: Clean the outer surface of the sensor with tap water. If there is still debris left, wipe it with a soft, damp cloth. For some stubborn dirt, add some household washing liquid to the tap water to clean it.
- Check the cable of the sensor: the cable should not be tight when working normally, otherwise the cable inside the cable will be broken and the sensor will not work properly.
- Check the sensor's measurement window for dirt.

1.3 Notes

The probe contains sensitive optical and electronic components. Make sure the probe is not subject to severe mechanical shock. There are no parts inside the probe that require user maintenance.

2. Sensor calibration

a) Zero calibration: use a large amount of beaker to take a proper amount of zero suspension or a known concentration of suspension (0 ~ 200.0mg / L), the sensor is placed vertically in the solution, the front end of the sensor is at least 10cm from the bottom of the beaker, 1 minute Zero calibration is performed after the value is stable. The instructions refer to the appendix.

b) Slope calibration: Place the sensor probe in a suspension of known concentration (200.0~2000.0mg/L). The front end of the sensor is at least 10cm away from the bottom of the beaker. After 1 minute, the value is stabilized and the slope is calibrated. The instructions refer to the appendix.

Note: No bubbles can be generated at the front of the probe.

3. Frequently Asked Questions

Problem	Possible reason	Solution
The operation interface cannot be connected or the measurement result is not displayed.	Controller and cable connection error	Reconnect controller and cable
	Cable failure	Please contact us
The measured value is too high, too low or the value is continuously unstable.	The sensor window is attached by a foreign object	Cleaning the sensor window surface

V. Quality and service

1. Quality assurance

- The quality inspection department has standardized inspection procedures, advanced and perfect testing equipment and means, and strictly in accordance with the regulations, to do 72-hour aging test and stability test on the product, and not to allow one unqualified product to leave the factory.
- The receiving party directly returns the product batch with a failure rate of 2%, and all the costs incurred are borne by the supplier. The reference standard refers to the product description provided by the supplier.
- Guarantee the quantity of goods and the speed of shipment.

2. Accessories and spare parts

This product includes:

- 1 sensor
- 1 cable (5 meters)

3. After-sales service commitment

The company provides local after-sales service within one year from the date of sale, but does not include damage caused by improper use. If repair or adjustment is required, please return it, but the shipping cost must be conceited. Damaged on the way, the company will repair the damage of the instrument for free.

Appendix data communication

1. Data format

The default data format for Modbus communication is: 9600, n, 8, 1 (baud rate 9600bps, 1 start bit, 8 data bits, no parity, 1 stop bit).

2. Information frame format

a) Read data instruction frame

0E	03	xx xx	xx xx	xx xx
Address	Function code	Register address	Number of registers	CRC check code (low byte first)

b) Read data response frame

0E	03	xx	xx.....xx	xx xx
Address	Function code	Bytes	Answer data	CRC check code (low byte first)

c) Write data instruction frame

0E	06	xx xx	xx xx	xx xx
Address	Function code	Register address	Write data	CRC check code(low byte first)

d) Write data response frame (same data command frame)

0E	06	xx xx	xx xx	xx xx
Address	Function code	Register address	Write data	CRC check code (low byte first)

3. Register address

Register address	Name	Instruction	Number of registers	Access method
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40001 (0x0000)	Measured value + temperature	Four double-byte integers are measured, measured decimal and temperature decimal places.	4 (8 bytes)	Read
44097 (0x1000)	Zero calibration	Calibrate in zero suspension, write data is 0; The read out data is the zero offset. (Calibration can also be carried out in the suspension of 0-200.0mg/L, and the calibration method refers to slope calibration).	1 (2 bytes)	Write/Read
44101 (0x1004)	Slope calibration	In the known suspension (200.0mg/L-2000.0mg/L), the written data is the actual value of the standard solution $\times 10$, and the readout data is the slope value $\times 1000$.	1 (2 bytes)	Write/Read
44113 (0x1010)	Temperature calibration	In the solution, the written data is the actual temperature value $\times 10$, and the readout data is the temperature calibration offset $\times 10$.	1 (2 bytes)	Write/Read
48195 (0x2002)	Sensor address	The default is 14, and the write data range is 1 - 255.	1 (2 bytes)	Write/Read
48196 (0x2003)	Baud rate	The default is 9600. 0--9600, 1--19200.	1 (2 bytes)	Write/Read
44865 (0x1300)	Automatic cleaning interval setting	The default is 30 minutes, the data range is 6-6000 minutes.	1 (2 bytes)	Write/Read
44866 (0x1301)	Automatic cleaning circle setting	The default is 3 circles, the data range is 0-6 circles.	1 (2 bytes)	Write/Read

48225 (0x2020)	Reset sensor	The calibration value is restored to the default value and the write data is 0. Note that the sensor needs to be re-calibrated after it has been reset before it can be used.	1 (2 bytes)	Write
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4. Command example

a) Start measurement command

Function: Get the suspended solids value and temperature value measured by the sensor; the unit of temperature is °C and the unit of suspended solids is mg/L.

Request frame: 0E 03 00 00 00 04 44 F6

Response frame: 0E 03 08 01 02 00 01 00 B0 00 01 BA 28

Reading example:

Suspended matter value	Temperature value
01 02 00 01	00 B0 00 01

For example: the suspended solids value 01 02 represents the suspended solids value in hexadecimal reading, 00 01 represents the suspended solids value with 1 decimal point, and the converted decimal value is 25.8.

The temperature value 00 B0 represents the hexadecimal reading temperature value, and 00 01 represents the temperature value with 1 decimal point converted into a decimal value of 17.6.

b) Calibration instructions

Zero point calibration

Function: Set the zero point calibration value of the suspended solids sensor; here the zero point calibration is performed in zero suspended solids liquid;

Request frame: 0E 06 10 00 00 00 8D F5

Response frame: 0E 06 10 00 00 00 8D F5

slope calibration

Function: Set the slope calibration value of the suspended solids sensor; the slope value here is based on the actual standard solution value x10, take 1000mg/L as an example to refer to the calibration;

Request frame: 0E 06 10 04 27 10 D6 08

Response frame: 0E 06 10 04 27 10 D6 08

c) Set device ID address:

Function: Set the Modbus device address of the sensor;

Change the sensor address 0E to 01, the example is as follows

Request frame: 0E 06 20 02 00 01 E2 F5

Response frame: 0E 06 20 02 00 01 E2 F5

5. Error response

If the sensor does not correctly execute the host command, it will return the following format information:

Definition	Address	Function code	Code	CRC check
Data	ADDR	COM+80H	xx	CRC 16
Number of bytes	1	1	1	2

- a) CODE: 01 –Function code error
03 – Data is wrong
b) COM: The received function code